

June 17, 2026

July 31, 2026

University of Southampton, Southampton, United Kingdom.

Dear PhD Admissions Committee,

I am excited to apply for the Quantum Reservoir Computing for photonic and quantum materials modelling PhD position at the University of Southampton. With a strong foundation in AI Data Science Engineering, I am confident that my skills and experience align perfectly with the requirements of this project.

As an AI Data Science Engineer, I have had the opportunity to work on various projects that leverage machine learning and high-performance computing. One such project that stands out is *MedicalMambaTS, a linear-time state space model for sub-millisecond bedside inference. In this project, I utilized PyTorch Autograd to develop a Physics-Informed Operator Network (PI-DeepONet) that leverages the dynamics of physical systems to achieve efficient*

I am particularly excited about the prospect of applying QRC as a new computational paradigm for modelling and designing complex quantum and photonic systems. My background in machine learning and high-performance computing, combined with my experience developing Physics-Informed Operator Networks, makes me well-suited to tackle the challenges of this project.

I believe that my skills and experience make me an ideal candidate for this PhD position. I am confident that my contributions will be valuable to the project, and I look forward to the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui